Fundamentals. Discrete compounding: $\prod_{j=0}^{n-1}(1+r_j)$. Continuous: $\exp(\int_0^T r_u du)$. Discrete discount: $\prod_{j=0}^{n-1}(1+r_j)^{-1}$. Continuous: $\exp(-\int_0^T r_u du)$. Discount factor: $P(t,T) = \mathbb{E}_t^Q[\exp(-\int_t^T r_u du)]$. Properties: $0 < P(t,T) < 1$, decreasing & convex in T. ZC yield: $zcy(t,T) = -\frac{1}{T-t}\ln P(t,T)$. Forward rates (simple): $F(T_1,T_2) = \frac{1}{T_2-T_1}(\frac{1+i(T_2)T_2}{1+i(T_1)T_1} - 1)$. Continuous: $F(T_1,T_2) = \frac{1}{T_2-T_1}\ln\frac{P(0,T_1)}{P(0,T_2)}$. From ZCY: $F(T_1,T_2) = \frac{zcy(0,T_2)T_2 - zcy(0,T_1)T_1}{T_2-T_1}$. Inst. fwd rate: $f(T) = \lim_{T_2\to T} F(T,T_2) = -\frac{\partial \ln P(0,T)}{\partial T}$. Relation: $P(0,T) = \exp(-\int_0^T f(u)du)$. Money mkt instruments: short maturity (¡1yr), no interim payments (T-bills, commercial papers, banker's acceptances, munic. notes). Discount basis: $P = M(1-i(T)T)$, yield $y_d = \frac{M-P}{MT}$ [N.Am., UK, EU]. Money-mkt basis: $P = \frac{M}{1+i(T)T}$, yield $y_m = \frac{M-P}{PT}$ [Japan]. Repos: sell securities w/ repurchase commitment.

Bonds. ZC bond: Value $P(0,T)$ for unit face at maturity T. Coupon-bearing (par, semi-annual): $\sum_{j=1}^{2T}\frac{c}{2}MP(0,\frac{j}{2}) + MP(0,T)$ [c=coupon rate, M=face, T=maturity]. Embedded options: callable (issuer redeems), convertible (holder converts to equity), putable (holder redeems).

FRA. OTC, notional M, effective T_1, termination T_2, ref. rate i_t, fixed R. Buyer CF at T_1: $\frac{M(T_2-T_1)(i_{T_1}(T_2)-R)}{1+i_{T_1}(T_2)(T_2-T_1)}$. If initial value nil: $R = F(T_1,T_2)$. Buyer borrows at R for $[T_1,T_2]$.

FRA Futures. Montreal: CORRA (1mo, 3mo), quoted $100-R$, size $\$2500\times(100-R)$ [\$25/bp]. CME: SOFR (1mo, 3mo), up to 10yr. Hedging: sell h futures (3mo, M mil. loan variable rate): $h = \frac{M\text{ mil.}\times 1\text{ bp}\times 91/365}{25} \approx M$.

T-Bill Futures. CME: quoted $100-R$ (13wk T-bill yield), size $\$2500\times(100-R)$ [\$25/bp], 1yr traded ref. quarters.

T-Bond Futures. Montreal: 4 maturities (2.5,10,20,30yr), \$100k nominal, 6% coupon, CF adjusts delivered bond to 6%. Delivery options: timing (any day in delivery month), end-of-month (days between last trading & delivery), quality (choose bond from pool). Invoice price: $IP_t = H_{t,T}\times CF$. Forward price (cash & carry): $FP_{t,T} = (SP_t + AC_t)(1 + r\frac{T-t}{365}) - AC_T - \frac{c}{2}(1 + r\frac{T-s}{365})$ [SP_t=spot, AC_t=accrued int., s=last coupon before T]. Basis after carry: $BAC_t = FP_{t,T} - IP_t$. CTD bond: minimize BAC_t (maximize r^*). Implied repo rate: $r^* = \frac{H_{t,T}\times CF - SP_t + AC_T - AC_t - c/2}{(SP_t+AC_t)\frac{T-t}{365} - \frac{c}{2}\frac{T-s}{365}}$.

Options on Bond Futures. American on bond futures (Montreal: 2,5,10yr), maturities 3mo preceding futures. Weekly also avail.

Caps & Floors. OTC. Cap=chain of Eur. calls ("caplets"). Floor=chain of Eur. puts ("floorlets"). Contract: notional M, ref. i_t, strike K, maturity T, settlements $t_1,\dots,t_n = T$. Period $\tau = t_j - t_{j-1}$ (typically qtly). Caplet payoff at t_j: $M\tau(i_{t_{j-1}}(t_j) - K)^+$. This is put on $P(t_{j-1},t_j)$ w/ strike $K' = \frac{1}{1+K\tau}$, nominal $M' = M(1+K\tau)$, maturity t_{j-1}. $\text{Cap}(M,K) = \sum_{j=\alpha+1}^n M'\text{Put}(P(t_{j-1},t_j),K')$. $\text{Floor}(M,K) = \sum_{j=\alpha+1}^n M'\text{Call}(P(t_{j-1},t_j),K')$ [t_α=first reset, $t_{\alpha+1}$=first settlement].

Swaps. Collar: Buy cap at K_2, sell floor at K_1 ($K_1 < K_2$). Net at t_j: $-M\tau i_{t_{j-1}}(t_j) + M\tau(i_{t_{j-1}}(t_j) - K_2)^+ - M\tau(K_1 - i_{t_{j-1}}(t_j))^+$. When $K_1 = K_2 = K$, collar=swap. No-arb: $\text{Cap}(M,i,K) - \text{Floor}(M,i,K) = \text{Payer swap}(M,i,K)$. Payer swap (pays fixed K, receives floating): $\text{PS}(M,i,K) = M\tau\sum_{j=\alpha+1}^n P(0,t_j)(F(t_{j-1},t_j) - K)$ where $F(t_{j-1},t_j) = \frac{1}{\tau}(\frac{P(0,t_{j-1})}{P(0,t_j)} - 1)$. Simplified: $\text{PS}(M,i,K) = M(P(0,t_\alpha) - P(0,t_n)) - M\tau K\sum_{j=\alpha+1}^n P(0,t_j)$. Receiver swap: $\text{RS}(M,i,K) = M(P(0,t_n) - P(0,t_\alpha)) + M\tau K\sum_{j=\alpha+1}^n P(0,t_j)$. Fwd swap rate: $K_f = \frac{P(0,t_\alpha)-P(0,t_n)}{\tau\sum_{j=\alpha+1}^n P(0,t_j)}$ (zero value for swaps).

Swaptions. Eur. opt. on swap. Payer swaption: right to enter payer swap. Receiver: right to enter receiver. Maturity=t_α, tenor=$t_n - t_\alpha$. Payer payoff: $M\tau\sum_{j=\alpha+1}^n P(t_\alpha,t_j)(F_{t_\alpha}(t_{j-1},t_j) - K)^+$. Swaption=opt. on coupon bond. Swaption value < cap/floor: $(\sum_{j=\alpha+1}^n P(t_\alpha,t_j)(F_{t_\alpha}(t_{j-1},t_j) - K))^+ \le \sum_{j=\alpha+1}^n P(t_\alpha,t_j)(F_{t_\alpha}(t_{j-1},t_j) - K)^+$.

Note: Jan(31), Feb(28), Mar(31), Apr(30), May(31), Jun(30), Jul(31), Aug(31), Sep(30), Oct(31), Nov(30), Dec(31).

Yield Curve Smoothing. Money mkt to ZC: For t-maturity money mkt rate $R_M(t)$ (Actual/365 Canada, Actual/360 US): $P(0,t) = 1 - R_M(t)\cdot t$. Bootstrapping: Long-term swap/bond rates $R_S(T)$ (bullet, semi-annual): $100 = \sum_{i=1}^T \frac{100R_S(T)}{2}P(0,\frac{i}{2}) + 100P(0,T)$. Solve recursively: $P(0,1/2)$, $P(0,1)$ from money mkt. From 2yr rate $R_S(2)$, solve $P(0,2)$. For $P(0,1.5)$, interpolate linearly between $P(0,1)$ and $P(0,2)$. **Nelson-Siegel-Svensson:** $\text{zcy}_{\text{NSS}}(T) = a + b(\frac{1-e^{-T/\tau}}{T/\tau}) + c(\frac{1-e^{-T/\tau}}{T/\tau} - e^{-T/\tau}) + d(\frac{1-e^{-T/\theta}}{T/\theta} - e^{-T/\theta})$. Params: a=long-end level, b=short-end/slope, c,d=curvatures, τ,θ=curvature maturities. Inst. fwd: $f_{\text{NSS}}(T) = a + \frac{e^{-T/\tau}}{T}(b\tau + cT) + \frac{e^{-T/\theta}}{T}dT$. Fitting: $\min_{a,b,c,d,\tau,\theta}\sum_{i=1}^n(\text{zcy}_{\text{NSS}}(t_i) - \text{zcy}(t_i))^2 w_i^2$. Weights: $w_i = 1$ (none), $w_i = \frac{1/t_i}{\sum_k 1/k}$ (short-end), $w_i = \frac{i}{\sum_k k}$ (long-end).

One-Factor Models. Vasicek (equilibrium). EMM dynamics: $dr_t = \kappa(\theta - r_t)dt + \sigma dW_t$ [κ=speed mean reversion, θ=long-run mean, σ=vol, all > 0]. Neg. rates possible. $r_T = r_t e^{-\kappa(T-t)} + \theta(1 - e^{-\kappa(T-t)}) + \sigma\int_t^T e^{-\kappa(T-u)}dW_u$. $\mathbb{E}_t[r_T] = \theta + (r_t - \theta)e^{-\kappa(T-t)}$. $\text{Var}_t[r_T] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa(T-t)})$. Physical: $dr_t = \kappa^P(\theta^P - r_t)dt + \sigma dZ_t$ where $\kappa^P = \kappa - \sigma\lambda_1$, $\theta^P = \frac{\kappa\theta + \sigma\lambda_0}{\kappa - \sigma\lambda_1}$. Stationarity: $\lambda_1 < \kappa/\sigma$. $P_{vk}(t,T) = a_{vk}(t,T)\exp[-b_{vk}(t,T)r_t]$, $\text{zcy}_k(t,T) = -\frac{1}{T-t}\ln[a_{vk}(t,T)) - b_{vk}(t,T)r_t]$ where $a_{vk}(t,T) = \exp[(\theta - \frac{\sigma^2}{2\kappa^2})(b_{vk}(t,T) - T + t) - \frac{\sigma^2}{4\kappa}b_{vk}(t,T)^2]$, $b_{vk}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$. Eur. call on \$1 ZC bond (maturity T, strike K, opt. maturity T_{call}): $\text{call}_{vk}(P,K) = P_{vk}(0,T)\Phi(h) - KP_{vk}(0,T_{\text{call}})\Phi(h - \sigma_p)$, $h = \frac{\sigma_p}{2} + \frac{1}{\sigma_p}\ln\frac{P_{vk}(0,T)}{KP_{vk}(0,T_{\text{call}})}$, $\sigma_p = \frac{\sigma}{\kappa}(1 - e^{-\kappa(T-T_{\text{call}})})\sqrt{\frac{1-e^{-2\kappa(T_{\text{call}}-t)}}{2\kappa}}$. $\text{put}_{vk}(P,K) = KP_{vk}(0,T_{\text{put}})\Phi(\sigma_p - h) - P_{vk}(0,T)\Phi(-h)$. Interpretation: Bond value is log-normally distributed. σ_p=volatility of ZC bond returns; $\Phi(h)$=call delta; $\Phi(h-\sigma_p)$=probability (under EMM) call is exercised. Opt. on coupon bond: $P_c(0,\{T_i\}) = \sum_{i=1}^n c_i P(0,T_i)$. Eur. call (strike K, maturity T_{call}) payoff: $(\sum_{i=1}^n c_i P(T_{\text{call}},T_i) - K)^+$. Jamshidian decomp: Find r^* where $\sum_{i=1}^n c_i P(T_{\text{call}},T_i,r^*) = K$. Then $\text{call}(P_c(0,\{T_i\}),K) = \sum_{i=1}^n c_i\text{call}(P(0,T_i),K_i)$ where $K_i = P(T_{\text{call}},T_i,r^*)$. Swaptions: Payer swaption=right to enter payer swap (fixed K, first reset t_α, settlements $t_{\alpha+1},\dots,t_n$). Pseudo-coupon bond: $c_i = K\tau$ for $i = \alpha+1,\dots,n-1$; $c_n = 1 + K\tau$. Find r^* (ATM): $\sum_{i=\alpha+1}^n c_i P(t_\alpha,t_i,r^*) = 1$. payer swaption$(M,K,t_\alpha) = M\sum_{i=\alpha+1}^n c_i\text{put}(P(t_\alpha,t_i),K_i)$. receiver swaption$(M,K,t_\alpha) = M\sum_{i=\alpha+1}^n c_i\text{call}(P(t_\alpha,t_i),K_i)$ where $K_i = P(t_\alpha,t_i,r^*)$.

CIR (equilibrium). EMM: $dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$. Params as Vasicek. Square-root ensures $r_t \ge 0$ if $r_0 \ge 0$ & Feller: $2\kappa\theta > \sigma^2$. Distribution: $r_t = Y/b_t$ where $b_t = \frac{4\kappa}{\sigma^2(1-e^{-\kappa t})}$, $Y \sim \chi_\nu^2(\lambda_c)$ w/ $\nu = 4\kappa\theta/\sigma^2$ d.o.f., non-centrality $b_t r_0 e^{-\kappa t}$. Physical: $dr_t = \kappa^P(\theta^P - r_t)dt + \sigma\sqrt{r_t}dZ_t$ where $\kappa^P = \kappa - \sigma\lambda$, $\theta^P = \frac{\kappa\theta}{\kappa - \sigma\lambda}$. Stationarity: $\lambda < \kappa/\sigma$. ZC bond: $P_{\text{cir}}(t,T) = a_{\text{cir}}(t,T)\exp[-b_{\text{cir}}(t,T)r_t]$, $a_{\text{cir}}(t,T) = (\frac{2\gamma e^{\frac{\kappa+\gamma}{2}(T-t)}}{(\kappa+\gamma)(e^{\gamma(T-t)}-1)+2\gamma})^{\frac{2\kappa\theta}{\sigma^2}}$, $b_{\text{cir}}(t,T) = \frac{2(e^{\gamma(T-t)}-1)}{(\kappa+\gamma)(e^{\gamma(T-t)}-1)+2\gamma}$, $\gamma = \sqrt{\kappa^2 + 2\sigma^2}$. $\text{zcy}(t,T) = -\frac{1}{T-t}[\ln(a_{\text{cir}}(t,T)) - b_{\text{cir}}(t,T)r_t]$. Eur. call (strike K, maturity T_{call}): $\text{call}_{\text{cir}}(P,K) = P_{\text{cir}}(0,T)\chi^2(\frac{2\xi\ln(a_{\text{cir}}(T_{\text{call}},T)/K)}{b_{\text{cir}}(T_{\text{call}},T)}, \frac{4\kappa\theta}{\sigma^2}, 2\phi^2 r e^{\gamma T_{\text{call}}}) - KP_{\text{cir}}(0,T_{\text{call}})\chi^2(\frac{2\ln(a_{\text{cir}}(T_{\text{call}},T)/K)}{b_{\text{cir}}(T_{\text{call}},T)(\phi+\psi)}, \frac{4\kappa\theta}{\sigma^2}, \frac{2\phi^2 r e^{\gamma T_{\text{call}}}}{\phi+\psi})$ where $\phi = \frac{2\gamma}{\sigma^2(e^{\gamma T_{\text{call}}}-1)}$, $\psi = \frac{\gamma+\kappa}{\sigma^2}$, $\xi = \phi + \psi + b_{\text{cir}}(T_{\text{call}},T)$.

Hull-White (arbitrage). $dr_t = \kappa(\varphi(t) - r_t)dt + \sigma dW_t$. $\varphi(t)$ (deterministic) calibrates to observed curve at $t = 0$. At $t = 0$: $P_{\text{hw}}(0,T) = P^m(0,T)$ (observed). ZC bond: $P_{\text{hw}}(t,T) = a_{\text{hw}}(t,T)\exp[-b_{\text{hw}}(t,T)r_t]$ where $b_{\text{hw}}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$ (same as Vasicek), $a_{\text{hw}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\exp[b_{\text{hw}}(t,T)f^m(0,t) - \frac{\sigma^2}{4\kappa}b_{\text{hw}}(t,T)^2(1 - e^{-2\kappa t})]$. Other formulas follow Vasicek.

Brigo-Mercurio (arbitrage). Shifted CIR: $dx_t = \kappa(\theta - x_t)dt + \sigma\sqrt{x_t}dW_t$, $r_t = x_t + \varphi(t)$. Fits curve when $\varphi(t) = f^m(0,t) - f^{cir}(0,t)$. WARNING: doesn't guarantee positive rates. Positivity if $\varphi(t) > 0\,\forall t$. ZC bond: $P_{\text{bm}}(t,T) = a_{\text{bm}}(t,T)\exp[-b_{\text{bm}}(t,T)x_t]$ where $b_{\text{bm}}(t,T) = b_{\text{cir}}(t,T)$, $a_{\text{bm}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\frac{a_{\text{cir}}(0,t)e^{-b_{\text{cir}}(0,t)x_0}}{a_{\text{cir}}(0,T)e^{-b_{\text{cir}}(0,T)x_0}}a_{\text{cir}}(t,T)e^{\varphi(t)(T-t)}$. Options follow CIR.

Black-Karasinski (arbitrage). Log-normal: $d\ln(r_t) = \kappa(\varphi(t) - \ln(r_t))dt + \sigma dW_t$. r_t is log-normal & always positive. Let $x_t = \ln(r_t)$, then x_t follows Vasicek w/ mean-reverting level $\varphi(t)$. Thus $r_t = \exp(x_t)$. By Itô: $\frac{dr_t}{r_t} = (\kappa\varphi(t) - \kappa\ln(r_t) + \frac{\sigma^2}{2})dt + \sigma dW_t$. Solution: $r_t = \exp[\ln(r_s)e^{-\kappa(t-s)} + \kappa\int_s^t \varphi(u)e^{-\kappa(t-u)}du + \sigma\int_s^t e^{-\kappa(t-u)}dW_u]$.

Multifactor Models. Two-Factor Gaussian (Hull-White 1994/Brigo-Mercurio 2006). Factors: $dx_t = -ax_t dt + \sigma dW_{1t}$, $dy_t = -by_t dt + \eta dW_{2t}$ ($x_0 = y_0 = 0$, W_{1t}, W_{2t} corr. ρ). Short rate: $r_t = x_t + y_t + \varphi(t)$. ZC bond: $P_{2fg}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\exp(B(t,T))$, $B(t,T) = \frac{1}{2}[V(t,T) - V(0,T) + V(0,t)] - \frac{1-e^{-a(T-t)}}{a}x_t - \frac{1-e^{-b(T-t)}}{b}y_t$. Eur. call: $\text{call}_{2fg}(P,K) = P_{2fg}(0,T)\Phi(h) - KP_{2fg}(0,T_{\text{call}})\Phi(h - \sigma_p)$ where $h = \frac{\sigma_p}{2} + \frac{1}{\sigma_p}\ln\frac{P_{2fg}(0,T)}{KP_{2fg}(0,T_{\text{call}})}$, σ_p from $V(t,T)$. Two-factor captures level & slope better than one-factor.

Fong-Vasicek. Two-factor Vasicek w/ rate vol. as 2nd factor. EMM: $dr_t = \kappa(\theta - r_t)dt + \sqrt{v_t}dZ_{1t}$, $dv_t = \gamma(\bar{v} - v_t)dt + \xi\sqrt{v_t}dZ_{2t}$ (Z_{1t}, Z_{2t} corr. ρ). r_t=short rate, v_t=variance. ZC bond: $P_{fvk}(t,T) = \exp[-b_{fvk,1}(t,T)r_t + b_{fvk,2}(t,T)v_t + b_{fvk,3}(t,T)]$ w/ $b_{fvk,1}(t,T) = b_{vk}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$, $b_{fvk,2}(t,T)$, $b_{fvk,3}(t,T)$ via confluent hypergeometric funcs (Selby-Strickland 1995 recursive alg.). Stochastic vol improves fit vs constant-vol Vasicek.

Longstaff-Schwartz. CIR w/ stochastic vol. EMM: $dr_t = \mu_r(r_t,v_t)dt + a\sqrt{\frac{\beta r_t - v_t}{a(\beta-a)}}dZ_{1t} + \beta\sqrt{\frac{v_t - ar_t}{\beta(\beta-a)}}dZ_{2t}$, $dv_t = \mu_v(r_t,v_t)dt + a^2\sqrt{\frac{\beta r_t - v_t}{a(\beta-a)}}dZ_{1t} + \beta^2\sqrt{\frac{v_t - ar_t}{\beta(\beta-a)}}dZ_{2t}$ (Z_{1t}, Z_{2t} indep.). Vol constraint: $a\tau \le v_t \le b\tau$. β large relative to α for stochastic vol. ZC bond: $P_{ls}(t,T) = a_{ls,1}(t,T)^y a_{ls,2}(t,T)^z\exp[b_{ls,1}(t,T)r_t + b_{ls,2}(t,T)v_t + b_{ls,3}(t,T)]$. Eur. call: $\text{call}_{ls}(P,K) = P_{ls}(0,T)\chi^2(\cdot;\cdot) - KP_{ls}(0,T_{\text{call}})\chi^2(\cdot;\cdot)$.

Two-Factor CIR (Brigo-Mercurio). $dx_t = \kappa_1(\theta_1 - x_t)dt + \sigma_1\sqrt{x_t}dZ_{1t}$, $dy_t = \kappa_2(\theta_2 - y_t)dt + \sigma_2\sqrt{y_t}dZ_{2t}$ (Z_{1t}, Z_{2t} indep.). Feller: $2\kappa_i\theta_i \ge \sigma_i^2$ ($i = 1,2$) ensures $x_t, y_t \ge 0$. Short rate: $r_t = x_t + y_t$. Arb-free: $r_t = x_t + y_t + \varphi(t)$ w/ $\varphi(t) = f^m(0,t) - f_1^{cir}(0,t) - f_2^{cir}(0,t)$. ZC bond: $P_{2fcir}(t,T) = a_{cir,1}(t,T)a_{cir,2}(t,T)\exp[-b_{cir,1}(t,T)x_t - b_{cir,2}(t,T)y_t]$ w/ $a_{cir,i}(t,T)$, $b_{cir,i}(t,T)$ from one-factor CIR, params $\{\kappa_i,\theta_i,\sigma_i\}$ for $i = 1,2$.

N-Factor. Each factor: $dx_{it} = \kappa_i(\theta_i - x_{it})dt + \sigma_i\sqrt{x_{it}}dZ_{it}$ (indep. BM). Short rate: $r_t = \sum_{i=1}^N x_{it}$ (or $+\varphi(t)$ arb-free). ZC bond: product of N one-factor CIR bonds.

Affine Term Structure (Duffie-Kan). N factors $X_t = (X_{1t},\dots,X_{Nt})' \in \mathbb{R}^N$. EMM: $dX_t = (K\theta - KX_t)dt + S_t dW_t$ where K is $N\times N$, θ is $N\times 1$, W_t N-dim BM, S_t $N\times N$ diagonal w/ $[S_t]_{ii} = \sqrt{a_i + \beta_i' X_t}$ [$a_i \ge 0$, $\beta_i \in \mathbb{R}^N$]. Market price of risk: $\lambda_t = \lambda_1 S_t + S_t\lambda_2 X_t$ [$\lambda_1 \in \mathbb{R}^N$, λ_2 $N\times N$]. Linear short rate: $r_t = \delta_0 + \delta_1' X_t$ [$\delta_0 \in \mathbb{R}$, $\delta_1 \in \mathbb{R}^N$]. ZC bond (exponential-affine): $P_{aff}(t,T) = \exp[A(\tau) - B(\tau)'X_t]$ [$\tau = T - t$].

Implementation. Estimation: fit to historical curve (ex-post). Calibration: fit to contemporaneous data (yields+derivatives) for precise pricing. Procedure: equilibrium models fit to yield curve (min sq. errors); arbitrage models fit to yield then caps/floors or swaption prices. Check price fit, more stringent: check implied vols (using Black's) fit.

Black for Caps/Floors. Assumes fwd rates log-normal; ignores corr. between discount & payoff; future spot rate $\approx$ future fwd rate $l_{t_{j-1}}(t_j) =$

$F(t_{j-1},t_j)$. Cap (strike K, maturity t_n): $\text{cap}(t_n) = M\sum_{j=i+1}^{n}\tau P(0,t_j)[F(t_{j-1},t_j)\Phi(d_1) - K\Phi(d_2)]$ [τ=accrual (typ. 0.25 qtly), M=notional]. $d_1 = \frac{1}{v_j}[\ln\frac{F(t_{j-1},t_j)}{K} + \frac{v_j^2}{2}]$, $d_2 = d_1 - v_j$, $v_j = \sigma_{\text{cap}}\sqrt{t_{j-1}}$ [σ_{cap}=implied vol from mkt, varies by tenor; first reset $t_0 = 0.25$, $t_1 = 0.5$, etc.]. Floor: $\text{floor}(t_n) = M\sum_{j=i+1}^{n}\tau P(0,t_j)[K\Phi(-d_2) - F(t_{j-1},t_j)\Phi(-d_1)]$. ATM strike: $K = K_f$ (fwd swap rate) when cap/floor ATM: $K_f = \frac{P(0,t_a)-P(0,t_n)}{\sum_{j=i+1}^{n}\tau P(0,t_j)}$.

Black for Swaptions. Ignores corr. between discount & payoff; assumes fwd swap rates driftless GBM under EMM. Payer swaption (strike K, maturity t_n): $\text{payer swaption}(t_n) = M(K_f\Phi(d_1) - K\Phi(d_2))\sum_{j=a+1}^{n}\tau P(0,t_j)$. Receiver: $\text{receiver swaption}(t_n) = M(K\Phi(-d_2) - K_f\Phi(-d_1))\sum_{j=a+1}^{n}\tau P(0,t_j)$ where K_f=fwd swap rate. $d_1 = \frac{1}{v}[\ln\frac{K_f}{K} + \frac{v^2}{2}]$, $d_2 = \frac{1}{v}[\ln\frac{K_f}{K} - \frac{v^2}{2}]$, $v = \sigma_{\text{swapt}}\sqrt{t_a}$ [σ_{swapt}=implied swaption vol from mkt, t_a=swap start].

Kalman Filter. State-space: measurement eq. relates observed ZCY to hidden states; transition eq. describes state evolution; estimates params maximizing log-likelihood. Measurement: $ZCY_t = A + BX_t + \eta_t$ [A $N\times 1$, B $N\times K$, η_t $N\times 1$ errors w/ diag. cov. $H = \text{diag}(h_{t_1}^2,\ldots,h_{t_N}^2)$; variances may differ by maturity but constant over time; uncorrelated cross-sectionally]. Transition: $x_{i,t} = \mathbb{E}[x_{i,t}|x_{i,t-\delta}] + \sqrt{\text{Var}[x_{i,t}|x_{i,t-\delta}]}\xi_t$. Algorithm: loop over obs. dates: (1) predict next-period state estimators & error variance; (2) compute prediction error (observed-predicted yields) & its variance; (3) compute log-likelihood from prediction error; (4) update state estimators using prediction error & updated variance. Final params maximize total log-likelihood.

Diebold-Li. Dynamic Nelson-Siegel. Time-varying loadings $L_a = 1$, $L_b = \frac{1-e^{-\tau/\tau}}{T}$, $L_c = (\frac{1-e^{-\tau/\tau}}{T} - e^{-\tau/T})$ [param τ fixed]. Factor representation: $zcy_t(T) = a_tL_a + b_tL_b + c_tL_c$ [a_t=level, b_t=slope, c_t=curvature]. Forecasting: Set τ so curvature loading L_c max at 30mo. AR(1) for each factor: $\hat{x}_{t+h} = c_x + k_x\hat{x}_t$ for $x = \{a,b,c\}$. Param estimation: OLS regression of x_{t+h} on x_t estimates intercept c_x & slope k_x minimizing sq. residuals between observed future factor values & predicted values.

HJM & Market Models. HJM Framework (Heath-Jarrow-Morton 1992). Models entire yield curve via inst. fwd rates $f(t,T)$ rather than short rate r_t. Key: specifying fwd rate vol. function $\sigma(t,T)$ fully determines dynamics of whole yield curve. HJM drift condition: Under EMM, drift entirely pinned down by vol—no free drift params. Fwd rate dynamics: $df(t,T) = \sigma(t,T)\int_t^T\sigma(t,u)du\,dt + \sigma(t,T)dW_t$. Specifying $\sigma(t,T)$ characterizes model for all maturities simultaneously. Markov condition (Carverhill 1994): $r_t = f(t,t)$ is Markovian iff $\sigma(t,T) = h(t)\cdot g(T)$ (separable). Then HJM reduces to generalized Hull-White, implementable on recombining tree. Vol specs: Ritchken-Sankarasubramanian: $\sigma(t,T) = \psi_t\exp(-\kappa(T-t))$ (ψ_t stochastic), leads to 2-state Markov (r_t,ϕ_t) for tree/PDE. Li-Ritchken-Sankarasubramanian: $\sigma(t,T) = v_t^{\beta}\exp(-\kappa(T-t))$, $0\le\beta\le 1$, interpolates Gaussian ($\beta = 0$) to log-normal ($\beta = 1$). Mercurio-Moraleda/Moraleda-Vorst: polynomial-exponential forms generate humped term structure of vol (vol peaks at intermediate maturities)—stylized fact of cap implied vol surfaces. Former lacks closed-form ZCB; latter has closed-form ZCB.

Market Models (BGM/LMM). Motivation: Mkt quotes cap/floor/swaption as Black implied vols, but Black rests on invalid assumptions (ignores discount-payoff corr.; assumes driftless GBM for fwd rates under EMM). Market models rigorously justify Black via change of prob. measure. Key: Simply-compounded fwd rate $F(t;T_1,T_2)$ is ratio of traded assets $/P(t,T_2)$, so it's T_2-fwd martingale under T_2-fwd measure $\mathbb{Q}^{T_2}$. Under this measure can model as driftless GBM, exactly recovering Black caplet formula w/o approx.

Cap Mkt Model (BGM/Miltersen-Sandmann-Sondermann/Jamshidian 1997). Each fwd rate $F(t;T_{k-1},T_k)$ follows driftless GBM under own fwd measure $\mathbb{Q}^{T_k}$. Caplets priced exactly w/ Black; cap=sum Black caplets. Calibration: fit vol. function $\gamma(t,T_k)$ (e.g., parametric humped $\gamma = [\alpha+\beta(T_k-t)]e^{-\delta(T_k-t)} + \theta$) to mkt cap implied vols minimizing sq. pricing errors. Swaptions can't be priced analytically in cap mkt model $\Rightarrow$ use MC simulation. Under diff. fwd measure $\mathbb{Q}^{T_n}$, fwd rates acquire drift correction; simulation uses drift-adjusted dynamics via Euler scheme.

Swap Mkt Model (Jamshidian 1997). Fwd swap rate $K_f(t)$ is ratio of traded assets, hence martingale under swap measure (annuity as numéraire). Models $K_f(t)$ as driftless GBM $\Rightarrow$ Black swaption formula exact. Cap/floor mkt model & swap mkt model theoretically incompatible: can't simultaneously use Black for both caps & swaptions in single consistent model. Practitioners choose one & price other numerically.

Backward-Looking Rates (Post-LIBOR, Lyashenko-Mercurio 2019). W/ LIBOR end, caps/floors/swaptions reference compounded overnight rates set in arrears (SOFR, CORRA). Backward-looking: rate for $[T_1,T_2]$ not known until T_2. Key: Extend ZC bond def. beyond maturity: for $t > T$, set $P(t,T) = B(t)/B(T)$ (bank account ratio), remains valid numéraire. Backward-looking fwd rate $\tilde{F}(t;T_1,T_2)$ still T_2-martingale, giving formula analogous to fwd-looking case. Vol adjustment: Before T_1, $\tilde{F}(t;T_1,T_2)$ behaves like fwd-looking rate (same vol $\gamma(t,T_2)$). Between T_1 & T_2, compounding overnight rates induces linear vol decay: $\bar{\gamma}(t;T_1,T_2) = \min(1,\frac{T_2-t}{T_2-T_1})\gamma(t,T_2)$. Market model structure & measure-change drift corrections carry over unchanged.

Numerical Methods. Binomial Trees (Heston 1995). Discrete-time arb-free tree for one-factor where short rate follows random walk w/ mean reversion. Dynamics: $r_{t+\Delta t} = \alpha + \rho r_t + \sigma\varepsilon$. Connection to continuous: equiv. to Vasicek/CIR w/ $\kappa = 1-\rho$, $\theta = \alpha/\kappa$. No-arb valuation: $g(t) = e^{-r_t\Delta t}[\pi g_u(t+\Delta t) + (1-\pi)g_d(t+\Delta t)]$ w/ risk-neutral prob. $\pi = \frac{e^{r_t\Delta t}-d}{u-d}$, $u = \exp[(-\alpha-\rho r_t-\sigma)\Delta t]$, $d = \exp[(-\alpha-\rho r_t+\sigma)\Delta t]$. Ensures martingale: discounted value process is martingale under EMM.

Trinomial Trees (Hull-White). Richer dynamics, faster convergence; more complex but better for mean-reverting. Designed for Hull-White (adaptable for log-normal like Black-Karasinski). Underlying: $dx_t = -\kappa x_t dt + \sigma dW_t$. Normally distributed, mean-reverting to zero. Tree structure: Each time step Δt, discretized value $x_{i,j}$ (i=time, j=space) branches to 3 nodes w/ transition probs π_u,π_m,π_d: Up: $x_{i+1,j+1}$; Middle: $x_{i+1,j}$; Down: $x_{i+1,j-1}$. Prob. constraints & moment matching: $\pi_u+\pi_m+\pi_d = 1$, $\pi_u(x_{i,j}+\Delta x_j) + \pi_m x_{i,j} + \pi_d(x_{i,j}-\Delta x_j) = x_{i,j}e^{-\kappa\Delta t}$. Variance: $\pi_u(x_{i,j}+\Delta x_j)^2 + \pi_m x_{i,j}^2 + \pi_d(x_{i,j}-\Delta x_j)^2 = [x_{i,j}+\Delta x_j(\pi_u-\pi_d)]^2 + \sigma^2\frac{1-e^{-2\kappa\Delta t}}{2\kappa}$. Branching geometry: adapts in regions to improve mean reversion & reduce neg. rates. Process (A): normal branching, states $\{x_{i,j}-\Delta x, x_{i,j}, x_{i,j}+\Delta x\}$. Process (B): $x_{i,j}$ too high (pulls down), states $\{x_{i,j}-2\Delta x, x_{i,j}-\Delta x, x_{i,j}\}$. Process (C): $x_{i,j}$ too low (pulls up), states $\{x_{i,j}, x_{i,j}+\Delta x, x_{i,j}+2\Delta x\}$. Critical levels $j_{\min}$, $j_{\max}$ determine branching switches.

2nd Stage: Calibration via Arrow-Debreu. Calibrate tree nodes to match mkt ZC bond prices using Arrow-Debreu prices Q_{ij} (state prices: value of claim paying \$1 at node j, time i). ZC bond pricing constraint w/ shift param: $P_{NSS}(0,(i+1)\Delta t) = \sum_k Q_{ik}\exp(-\alpha_i\Delta t)\pi_k$. Solve for shift param α_i to match observed bond price $P_{NSS}(0,(i+1)\Delta t)$ from mkt. State price propagation: $Q_{(i+1)j} = \sum_k Q_{ik}\pi_{kj}\exp(-(\alpha_i+k\Delta x)\Delta t)$ where π_{kj}=transition prob. from state k to j. Black-Karasinski extension: For log-normal rates, apply $f^{-1}(\cdot) = \exp(\cdot)$ & solve α_i numerically (nonlinear). Key: Match each bond maturity sequentially; one α_i per time step (one-factor); ensure positivity if normal models; use Newton-Raphson for implicit solutions.

Finite Difference. More flexible than trees (accommodates wide variety of models) but slower convergence (requires discretization of entire state-space).

1-D Explicit Scheme. One-factor diffusion under EMM: $dr_t = \mu(t,r_t)dt + \sigma(t,r_t)dW_t$. PDE for security value: $\frac{\partial g(t,r_t)}{\partial t} + \frac{\partial g(t,r_t)}{\partial r_t}\mu(t,r_t) + \frac{1}{2}\frac{\partial^2 g(t,r_t)}{\partial r_t^2}\sigma(t,r_t)^2 - r_t g(t,r_t) = 0$. Grid: space step Δr, time step Δt. Nodes indexed by i (space, bottom to top), t (time, current to maturity T). Discretized Greeks: $\Delta_i^t, \Gamma_i^t, \theta_i^t$ (delta, gamma, theta at node i, time t). Drift $\mu(t,r_t)$, vol $\sigma(t,r_t)$ discretized as μ_i^t, σ_i^t. Discretized PDE: $\theta_i^t + \Delta_i^t\mu_i^t + \frac{1}{2}\Gamma_i^t(\sigma_i^t)^2 - i\Delta r g = 0$. Central diff. approx: $\Delta_i^t = \frac{g_{i+1}^t - g_{i-1}^t}{2\Delta r}$, $\Gamma_i^t = \frac{g_{i+1}^t - 2g_i^t + g_{i-1}^t}{(\Delta r)^2}$, $\theta_i^t = \frac{g_i^t - g_i^{t-1}}{\Delta t}$. Backward recursion: $g_i^{t-1} = A_i^t g_{i-1}^{t-1} + B_i^t g_i^{t-1} + C_i^t g_{i+1}^{t-1}$ w/ $v_1 = \frac{\Delta t}{(\Delta r)^2}$, $v_2 = \frac{\Delta t}{\Delta r}$, coefs: $A_i^t = \frac{1}{2}[(\sigma_i^t)^2 v_1 - \mu_i^t v_2]$, $B_i^t = 1 - (\sigma_i^t)^2 v_1 - i\Delta r\Delta t$, $C_i^t = \frac{1}{2}[\mu_i^t v_2 + (\sigma_i^t)^2 v_1]$. Properties: Sum $A_i^t + B_i^t + C_i^t = 1 - i\Delta r\Delta t$ captures 1-period discount; embeds martingale property. Weights A_i^t, B_i^t, C_i^t not probabilities, may lie outside $[0,1]$, potentially causing instability. Stability requires small v_1, v_2: discretize time sufficiently relative to space (horizontally stretched grid). Choice of $r_{\max}$: balance truncation bias (need $r_{\max}$ high) vs computational efficiency (higher $r_{\max}$ requires finer discretization). Boundary conditions: ZC bonds: $g = 1$ when $r = 0$; $g = 0$ when $r = r_{\max}$. Deep ITM: linear approx $g_{i_{\max}}^t = 2g_{i_{\max}-1}^t - g_{i_{\max}-2}^t$. Deep OTM: derivatives worthless. American opts: compute max of exercise & continuation values recursively at each node; use exercise value as boundary for deep ITM regions.

2-D Explicit Scheme. Two-factor diffusion under EMM: $dx_s = \mu(t,x_s,y_t)dt + \sigma(t,x_s,y_t)dW_s^x$, $dy_t = \mu(t,x_s,y_t)dt + \sigma(t,x_s,y_t)dW_s^y$ w/ $dW_s^x dW_s^y = \rho dt$ (corr. ρ). Contingent claim value $g(t,x_s,y_t)$ satisfies: $\frac{\partial g}{\partial t} + \frac{\partial g}{\partial x_s}\mu(t,x_s,y_t) + \frac{1}{2}\frac{\partial^2 g}{\partial x_s^2}\sigma(t,x_s,y_t)^2 + \frac{\partial g}{\partial y_t}\mu(t,x_s,y_t) + \frac{1}{2}\frac{\partial^2 g}{\partial y_t^2}\sigma(t,x_s,y_t)^2 + \rho\sigma_x(t,x_s,y_t)\sigma_y(t,x_s,y_t)\frac{\partial^2 g}{\partial x_s\partial y_t} - r_t g(t,x_s,y_t) = 0$. 3-D grid: space steps Δx, Δy; time step Δt. Notation: $g_{i,j}^t$=value at node i for x (left to right), node j for y (bottom to top), time t. Discretization: Apply central diffs for partial derivs & backward recursion as 1-D case. Cross-partial deriv $\frac{\partial^2 g}{\partial x_s\partial y_t}$ discretized using 4-point stencil. Solution proceeds backward through time on 2-D grid at each time step.

Monte Carlo. Generating normal: Box-Muller: given u_1,u_2 uniform $[0,1]$: $\varepsilon_1 = \sqrt{-2\ln(u_1)}\cos(2\pi u_2)$, $\varepsilon_2 = \sqrt{-2\ln(u_1)}\sin(2\pi u_2)$. Euler discretization: $r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t$. Full truncation: replace r_t w/ $\max(r_t,0)$ in drift & diffusion to preserve non-negativity. Milstein: higher accuracy: $r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t + \frac{\sigma}{4}\Delta t(\varepsilon_t^2 - 1)$. Path-dependent valuation: For each path j, compute discount: $D^{(j)} = \exp(-\sum_{n=0}^{N-1} r_n^{(j)}\Delta t)$. Price security: Price $= \mathbb{E}^{\mathcal{Q}}[D^{(j)}\cdot X(T)] = \frac{1}{M}\sum_{j=1}^{M} D^{(j)}\cdot X^{(j)}(T)$ where $X(T)$=terminal payoff, M=# paths. Correlated BMs: Generate indep. $(\varepsilon_1,\varepsilon_2)$, then $\Delta W_1 = \sqrt{\Delta t}\varepsilon_1$, $\Delta W_2 = \sqrt{\Delta t}(\rho\varepsilon_1 + \sqrt{1-\rho^2}\varepsilon_2)$.

Path-Dependent Payoffs. Barrier caps/floors: Activated ("in") or deactivated ("out") first time ref. rate reaches critical barrier from below ("up") or above ("down"). Four types: up-in cap, down-in floor, down-out cap, up-out floor. Payoff: $(i_{u-1}^{(u)} - K)^+\cdot\mathbb{1}(\max_{s\in[u-1,u]} i_s^{(u)} > H)$. Barrier condition applies to whole cap/floor: once caplet activates, all subsequent activate. Bounded caps/floors: Total accumulated payout can't exceed M. At each settlement, accumulated value computed & test applied to determine if below limit: $(i_{u-1}^{(u)} - K)^+\cdot\mathbb{1}(\sum_{j=1}^{u-1}(i_{j-1}^{(j)} - K)^+ < M)$. Option deactivated first time test fails. Incremental fixed swaps: Fixed leg as weighted avg of fixed & variable rates. Payer swap (notional M, fixed K, freq τ): $-M\tau(\kappa + (1-x)u_t^{(u)}) + M\tau d_t^{(u)}$. Fraction x determined by ref. rate tier: $x = 100\%$ if $i_t < i_1$; $x = 60\%$ if $i_2 \le i_t \le i_1$; $x = 0\%$ if $i_t > i_2$. At each reset, simulated rate determines x & payoff. Ratchet caps/floors: Strike adjusts to level of ref. rate on preceding reset date. Payoff at settlement: $(i_{u-1}^{(u)} - i_{u-2}^{(u-1)})^+$. Simulation adjusted to compute diff between current & previous reset rates at each settlement. Rolling caps/floors: Nominal not constant. When caplet (floorlet) not exercised (not ITM), its notional added to next. At each reset, apply test & adjust payoff.

Derivatives Basics. Itô's product: $dW(t)dW(t) = dt$, $dt\,dt = 0$, $dW(t)dt = 0$, $dW_1(t)dW_2(t) = \rho dt$. **Itô's Lemma:** $df(t) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX(t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}[dX(t)]^2$. **Black-Scholes-Merton** (Eur. opts): $c(t,T,K) = \hat{S}(t)\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$, $p(t,T,K) = Ke^{-r(T-t)}\Phi(-d_2) - \hat{S}(t)\Phi(-d_1)$ where $d_1 = \frac{\ln(\hat{S}(t)/K) + (r+\frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$, $d_2 = d_1 - \sigma\sqrt{T-t}$. Discrete dividends: $\hat{S}(t) = S(t) - \sum_{i=1}^{n} d(t_i)e^{-r(t_i-t)}$. Continuous yield: $\hat{S}(t) = S(t)e^{-y(T-t)}$. **Pointers:** Normal PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp(-\frac{(x-\mu)^2}{2\sigma^2})$. $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. If $x \sim \mathcal{N}(\mu,\sigma^2)$ then $x = \mu + \sigma\epsilon$, $\epsilon \sim \mathcal{N}(0,1)$, $\mathbb{E}[e^x] = e^{\mu+\frac{1}{2}\sigma^2}$.